# Tutorial Questions

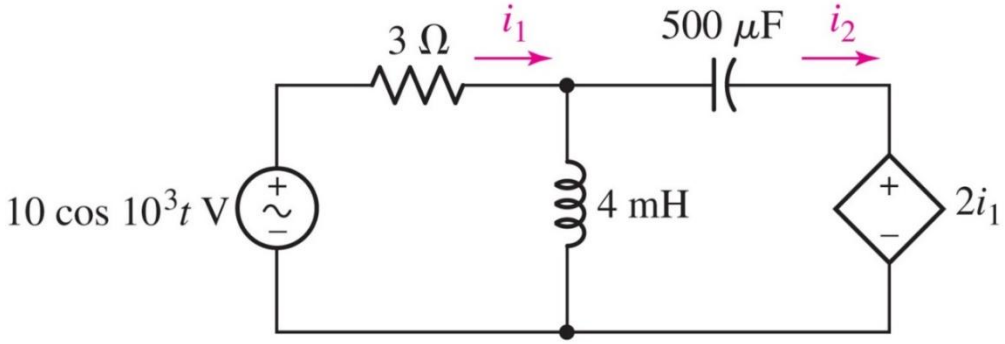
- For in-class discussion and reflection.

Week 9

# **Exercise 9.1:** Apply Mesh Analysis using Phasors

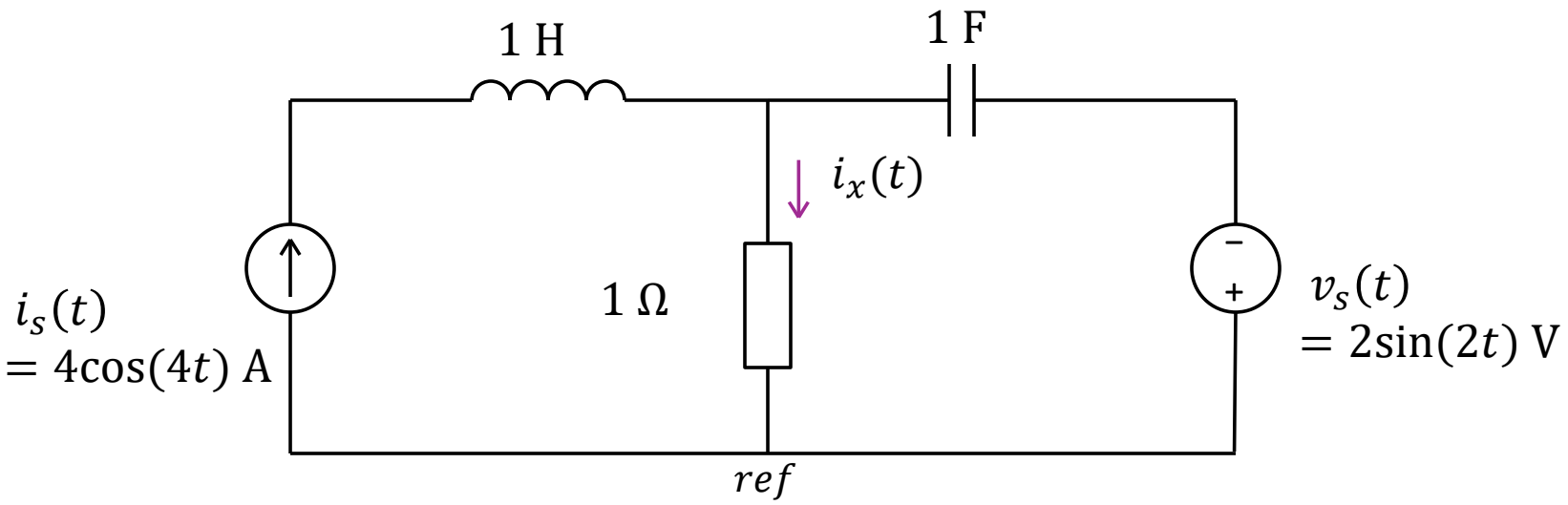


- Find the currents $i_1(t)$ and $i_2(t)$ through mesh analysis of phasors. ***Express the answers in both phasor/frequency and time domain.***



# **Exercise 9.2:** Superposition at different frequencies

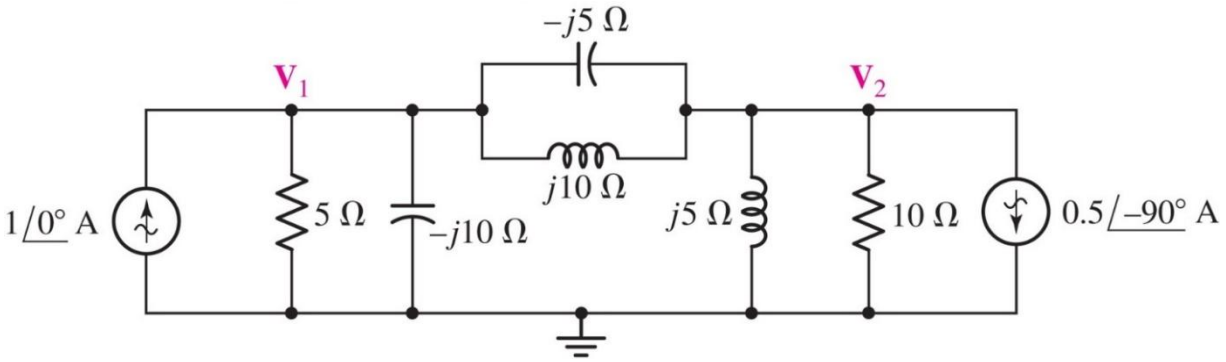
- Determine the downward current through the 1 Ω resistor using phasor analysis and superposition. Assume sinusoidal steady state.



# **Exercise 9.3:** Apply Thevenin's Theorem using Phasors



- Find $\mathbf{V}_1$ in the circuit below using Thevenin's equivalent circuit theorem.



# Self-practice Exercises

- to improve your true understanding and get your ready for final exam assessment.

Week 9

## Exercise questions for Part 3 content:

- There are four sources of questions for you to practice:
  - Lecture ppt's example questions, most with sample solutions.
  - Textbook questions in this Tutorial ppts as well as McGraw Hill Connect (LMO link), both with sample solutions
- Please be reminded that these questions are selected RANDOMLY (with absolutely no intention to relate to any assessment or exam) for self-practices. **They carry NO marks.**

**From CS Lim (Part 3)**

# Self practice (Page 1/3)

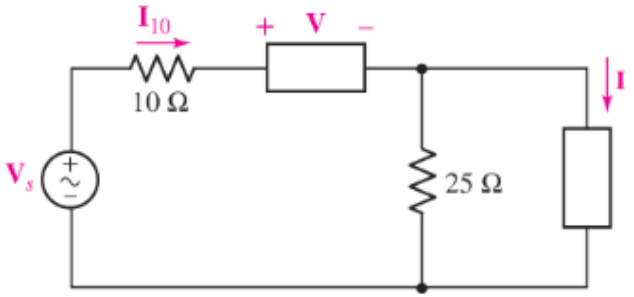



FIGURE 10.53

35. The circuit of **Fig. 10.53** is shown represented in the phasor (frequency) domain. If $\mathbf{I}_{10} = 4\angle 35^\circ$ A, $\mathbf{V} = 10\angle 35^\circ$ V, and $\mathbf{I} = 2\angle 35^\circ$ A, (*a*) across what type of element does $\mathbf{V}$ appear, and what is its value? (*b*) Determine the value of $\mathbf{V}_s$.

44. Employ phasor-based analysis to obtain an expression for $i(t)$ in the circuit of **Fig. 10.57**.

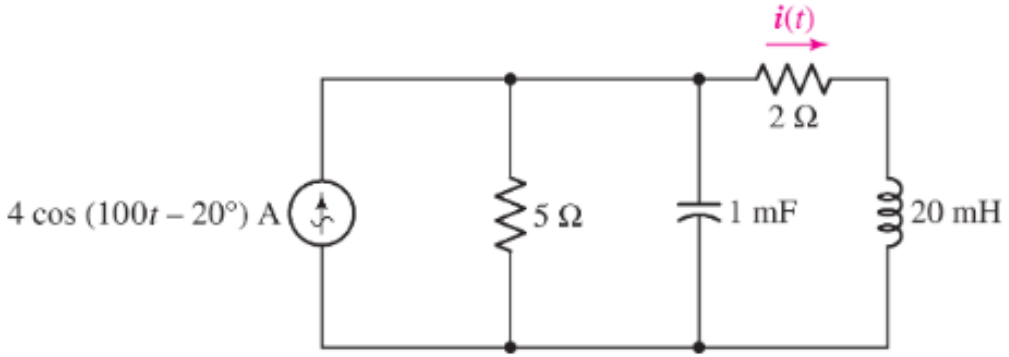



FIGURE 10.57

# Self practice (Page 2/3)

50. In the phasor-domain circuit represented by **Fig. 10.60**, let $\mathbf{V}_1 = 10\angle -80^\circ$ V, $\mathbf{V}_2 = 4\angle -0^\circ$ V, and $\mathbf{V}_3 = 2\angle -23^\circ$ V. Calculate $\mathbf{I}_1$ and $\mathbf{I}_2$.

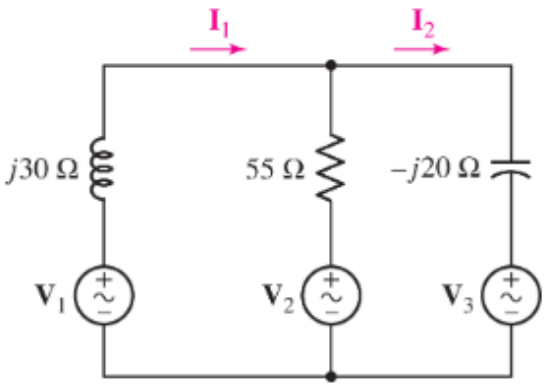



**FIGURE 10.60**

62. Determine $\mathbf{V}_1$ and $\mathbf{V}_2$ in **Fig. 10.68** if $\mathbf{I}_1 = 33\angle 3^\circ$ mA and $\mathbf{I}_2 = 51\angle -91^\circ$ mA.

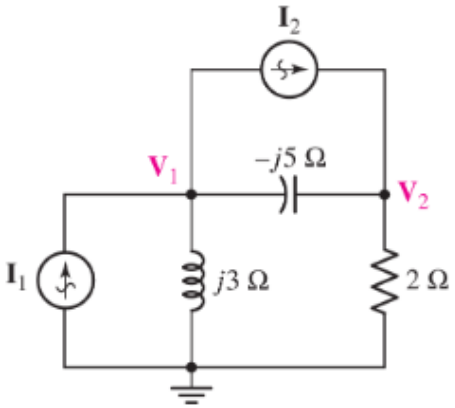



**FIGURE 10.68**

# Self practice (Page 3/3)

76. For the circuit shown in **Fig. 10.79**, (*a*) draw the phasor representation of the circuit; (*b*) determine the Thévenin equivalent seen by the capacitor, and use it to calculate $v_C(t)$. (*c*) Determine the current flowing out of the positive reference terminal of the voltage source. (*d*) Verify your solution with an appropriate LTspice simulation.

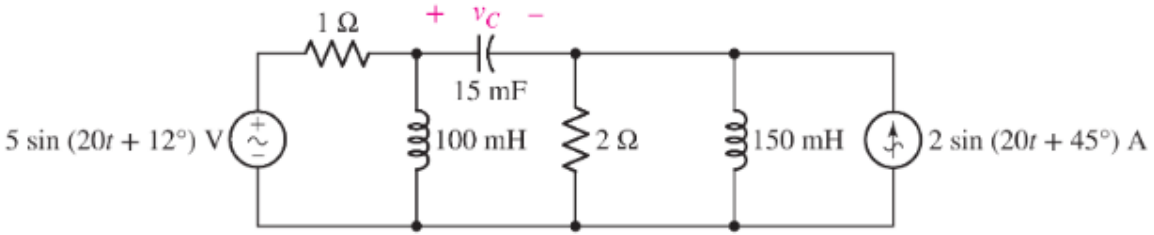



**FIGURE 10.79**